Moyan Liu

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RESEARCH FOCUS

Decision-making and control in complex adaptive systems, with a focus on how local instability and uncertainty can be leveraged to guide system evolution. My work applies this perspective to the Earth's climate system, combining dynamical systems theory, probabilistic modeling, and AI-based weather models to study the predictability and control of extreme events.

EDUCATION

Arizona State University, Tempe, AZ

Doctor of Philosophy in Complex Adaptive Systems Science (GPA: 4.0/4.0) Aug 2024 – Present

Supervisor: Prof. Upmanu Lall | Research Focus: Extreme Weather Control

Relevant Coursework: Space-Time Data Analysis, Cloud-Based Remote Sensing, Dynamical Systems and Dynamic Methods

Columbia University, New York, NY

Master of Science in Earth and Environmental Engineering (GPA: 3.7/4.0) Sep 2022 – Dec 2023

Relevant Coursework: Ocean and Atmosphere, GIS-Resources Environmental Infrastructure Management, Management and Development of Water Systems, Machine Learning for Environmental Engineer, Environmental Data Analysis

Southwest Minzu University, Chengdu, CN

Bachelor of Engineering in Environmental Engineering (GPA: 3.5/4.0) Sep 2018 – Jun 2022

Relevant Coursework: Advanced Mathematics, Probability & Mathematical Statistics, Environmental Engineering Principle

PUBLICATION

- **Liu, M.**, Huang, Q., & Lall, U. (2026). Instability-Aware Steering of an Extreme Atmospheric River in an AI Weather Foundation Model. *Geophysical Review Letter*, Under Review.
- **Liu, M.**, Huang, Q., & Lall, U. (2025). Regime identification and control of extremes in the nonautonomous Lorenz model with chaos and intransitivity. *Physical Review E*.
- **Liu, M.**, Huang, Q., & Lall, U. (2025). Targeted Adaptive Chaos Control of Regimes and Eddy Strength in Two Lorenz Models. *Chaos, Solitons & Fractals*.
- Huang, Q., **Liu, M.**, Kwon, Y, Kapnick, S& Lall, U. (2026). Steering Tropical Cyclones Through Targeted Thermodynamic Perturbations in an AI Weather Model. *Journal of Geophysical Research: Machine Learning and Computation*, Under Review.
- Huang, Q., **Liu, M.**, & Lall, U. (2025). Weather Jiu-Jitsu: Climate Adaptation for the 21st Century. *PLoS Water*.
- Huang, Q., **Liu, M.**, Kwon, Y& Lall, U. (2026). Evaluating the Predictability of Selected Weather Extremes with Aurora, an AI Weather Forecast Model. *Atmosphere*.

TECHNICAL SKILLS

Python (TensorFlow, PyTorch), R, ArcGIS Pro, Google Earth Engine, AutoCAD

RESEARCH EXPERIENCE

Latent State Representation in Weather Foundation Model, Arizona State University Feb 2026 – Present

- Extract and analyze encoder and transformer hidden states to identify latent feature representations of atmospheric circulation patterns.
- Develop reduced-order latent representations to construct lightweight surrogate models for efficient weather forecasting.

AI-Driven Diagnostics and Predictability of Extreme Weather Trajectories, Arizona State University July 2025 – present

- Develop AI-Based weather models for AR/TC tracking prediction and analyze extreme precipitation and flooding risks.
- Design trajectory control methods that optimize for minimal energy expenditure to enhance climate management efforts.

Dynamics and Predictability in Idealized Climate Models, Arizona State University July 2024 – Sep 2025

- Developed and implemented optimization methods in Lorenz systems to analyze trajectory evolution, transitions, and predictability.
- Applied local Lyapunov exponents and non-homogeneous hidden Markov models, to identify and anticipate regime shifts.

Wind Power Plant Site Selection in the United States using ArcGIS Pro, Columbia University Feb 2023 – July 2023

- Analyzed geospatial datasets in ArcGIS Pro to evaluate meteorological, geographic, and engineering constraints for wind farm siting.
- Developed Python-based ranking models to optimize sites, and validated results against existing infrastructure using suitability analysis.

Pocket Park Addressing Flood Issues in Mexico City, Columbia University Oct 2022 – Jan 2023

- Designed spatial layouts and sizing of pocket parks using ArcGIS Pro and AutoCAD to enhance urban flood resilience.
- Integrated sunken landscapes and decentralized water treatment systems, and quantified flood mitigation capacity with cost analysis.

Prediction of Flooding in Mexico City by Machine Learning, Columbia University Oct 2022 – Jan 2023

- Developed and implemented a Long Short-Term Memory (LSTM) model to forecast future flooding in Mexico City.
- Conducted a comparative analysis of neural network models (CNN, Random Forest) for flood prediction.

CONFERENCE PRESENTATIONS

- **M. Liu**, Q. Huang, and U. Lall. *Instability-Aware Perturbations of Extreme Events in an AI Weather Foundation Model*. Oral presentation at the EGU General Assembly 2026, Vienna, Austria.
- Q. Huang, **M. Liu**, and U. Lall. *Weather Jiu-Jitsu: Prospects for Atmospheric Nudging to Defuse the Impact of Catastrophic Weather Extremes*. Poster presentation at the EGU General Assembly 2026, Vienna, Austria.
- **M. Liu**, Q. Huang, and U. Lall. *Controlling Eddies in the Non-Autonomous Lorenz-84 Model*. Poster presentation at the APS Global Physics Summit, Denver, CO.
- **M. Liu**, Q. Huang, and U. Lall. *Weather Jiu-Jitsu: Toward Adaptive Control of Extreme Weather Regimes Using the Seasonally Forced Lorenz-84 Model*. Poster presentation at the AGU Fall Meeting 2025, New Orleans, LA.
- **M. Liu**, Q. Huang, U. Lall, Y. Kwon, and H.-H. Kwon. *Weather Jiu-Jitsu: Toward Controlling Atmospheric Rivers with Deep Learning Foundation Models and Adaptive Perturbation*. Poster presentation at the AGU Fall Meeting 2025, New Orleans, LA.
- Q. Huang, **M. Liu**, U. Lall, Y. Kwon, H.-H. Kwon, and Y. Liu. *Weather Jiu-Jitsu: Steering Tropical Cyclones with Targeted Perturbations in Deep Learning Models*. Poster presentation at the AGU Fall Meeting 2025, New Orleans, LA.
- Q. Huang, **M. Liu**, and U. Lall. *Weather Jiu-Jitsu: Climate Adaptation for the 21st Century*. Poster presentation at the AGU Fall Meeting 2025, New Orleans, LA.
- **M. Liu**, Q. Huang, and U. Lall. *Investigating Adaptive Chaos Control for Mitigating Weather Extremes*. Poster presentation at the AGU Fall Meeting 2024, Washington, DC.
- Q. Huang, **M. Liu**, and L. Josset. *Drinking Water Safety in NYS Schools: Progress, Challenges, and Data Gaps in Lead Testing and Reporting*. Poster presentation at the AGU Fall Meeting 2024, Washington, DC.

PROFESSIONAL EXPERIENCE

Technical and Performance Intern, Veolia North America, Paramus, NJ May 2024 – Aug 2024

- Developed machine learning models to optimize water treatment processes, reducing chemical and energy consumption.
- Assisted the team with data analysis for three case studies and drafted standard operating procedures (SOPs).

Data Analyst, Columbia Water Center, New York, NY Sep 2023 – July 2024

- Cleaned and analyzed drinking water quality and violation data from over 4,644 public schools in New York State.
- Visualized data for recommendations and submitted results to federal, state, and local stakeholders.

Research Assistant, Columbia Water Center, New York, NY July 2023 – Jan 2024

- Developed machine learning models to predict electricity prices using demand, generation mix, and climate variables.
- Designed risk management strategies, including insurance and derivative frameworks, to mitigate extreme price volatility.

RELEVANT EXPERIENCE

Vice President, Aquanauts ASU, Arizona State University, Tempe, AZ Jan 2025 – Present

- Coordinate technical discussions in project directions and organize collaboration across multiple stakeholders.
- Engage students, industry experts, and faculty from various disciplines to address water and climate issues.

Teaching Assistant, Columbia University, New York, NY Sep 2023 – Dec 2023
Course: 'Earth Resources and Sustainable Development'

- Led weekly discussion sections, guiding students in topics related to resource management, sustainability, and environmental systems.
- Evaluated student presentations and term papers, providing structured feedback to support analytical and research skill development.